

Peiyuan Chen

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Research Profile

Undergraduate physics student focusing on particle detection, PMT instrumentation, and data analysis for large-scale experiments such as PandaX.

Education

Jilin University

Bachelor of Science, Physics

Changchun, China
Sept 2020 – present

Georgia Institute of Technology

Visiting Honor Student, Physics

Atlanta, USA
Jan 2024 – May 2024

Research Experience

PandaX Collaboration, Shanghai Jiao Tong University, Research Assistant

Shanghai, China
June 2024 – present

- Contributed to R12699 PMT saturation and suppression measurements, waveform analysis, and charge reconstruction; co-first author of the resulting IEEE TNS paper.
- Led three students in diagnostics of a 1408-channel TPC readout system and built a database for shared measurements and fault tracking.
- Designed and implemented a 55-PMT readout system and studied charge-pattern position reconstruction using Geant4 simulations.
- Reported approximately 0.66 mm position resolution in the simulated configuration; not an experimentally measured full-detector resolution.
- Designed and tested flexible PMT base prototypes, identifying noise, crosstalk, high-voltage discharge, and mechanical limitations.

Georgia Institute of Technology, Research Intern (Trinity PeV Neutrino Observatory)

Atlanta, USA
Mar 2024 – May 2024

- Designed and fabricated detector connection components for the Trinity PeV Neutrino Observatory.

PandaX Collaboration, Research Assistant

Shanghai, China
Jan 2022 – Mar 2022

- Designed a calibration device for a high-pressure xenon detector with Micromegas readout.
- Reduced device thickness while addressing mechanical strength and leak-tightness requirements.

Publications

Performance Study and Circuit Simulation for R12699-406-M4 Photomultiplier Tube Base

June 2026

IEEE Transactions on Nuclear Science 73(6), 2511-2519. Published; co-first author with Houqi Huang.

Measurement of Solar pp Neutrino Flux with the New PandaX-4T Data

July 2026

arXiv:2607.02405. PandaX Collaboration paper; author list is ordered alphabetically.

Bi-Interlayer Strategy for Modulating NiCoP-Based Heterostructure toward High-Performance Aqueous Energy Storage Devices

July 2024

Advanced Materials. Co-author (fourth).

Manuscripts and Thesis

Light Source Position Reconstruction Using PMT Arrays in Liquid Xenon Detectors Undergraduate thesis.	2025
Comparative Study of Position Reconstruction Performance between Two-Inch and Three-Inch PMT Arrays In preparation; first author.	2025

Projects

3-Inch PMT Readout Array and Position Reconstruction
Built a 55-channel PMT readout system and developed reconstruction algorithms for large-scale xenon detectors

Flexible Base Circuit Design for PandaX-20T
Designed and characterized flexible PMT base prototypes; thermal cycling reached approximately -37 C, with remaining electrical and mechanical limitations.

PMT Linearity Characterization
Bench measurements and circuit simulation of PMT saturation and suppression; supports future correction studies.

Awards

National First Prize, China Undergraduate Physics Experiment Competition (CUPEC) Ranked 1st in comprehensive experimental problem D.	2023
High-Level Academic Competition Achievement Scholarship, Jilin University	2025
Student of the Year (Group Member), Shanghai Jiao Tong University	2023
13th Place, China Undergraduate Physics Tournament	2021
Grand Prize, Northeast Division Undergraduate Physics Tournament	2021

Patents

Small Wind Tunnel Device Used in Laboratory
Chinese patent CN215414302U.

Method of Measuring Wind Speed Using a Constant Flow Thermal Anemometer
Chinese patent CN113466487B.

Skills

Programming & Data: Python, C++, MATLAB, ROOT, Geant4, Git, LaTeX

Experimental: PMT characterization, detector readout, DAQ systems, circuit design, vacuum systems

Computation: Monte Carlo simulation, finite element analysis, machine learning for reconstruction

Languages

Chinese: Native

English: Fluent

Leadership and Service

CUPT Community, Jilin University , Founder & Chairman - Founded physics competition training community - Mentored students in experimental design and data analysis	Changchun, China Feb 2021 – June 2023
JLUPT , Chairman & Referee Organized and judged university physics tournament	Changchun, China May 2022 – May 2022